

Diachroneity of biostratigraphic events and paleoenvironmental reconstruction across a Northwest Pacific latitudinal transect



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Abstract

ODP Sites 1207, 1208, and 1209 provide a Northwest Pacific latitudinal transect for investigating Neogene–Quaternary calcareous plankton bioevent diachroneity and Kuroshio Current Extension (KCE) variability during the mid-Piacenzian Warm Period (mPWP). Nannofossil bioevents were compared with planktic foraminifera, while ~3-kyr stable-isotope and high-resolution XRF records were used to reconstruct paleoenvironmental variability. Both microfossil groups exhibit diachroneity, with event-specific magnitudes and latitudinal patterns; nannofossils generally show smaller offsets than planktic foraminifera. During the mPWP, thermocline conditions varied spatially and temporally, with changes in the direction and magnitude of the north–south temperature gradient. Integrated records reveal recurring cold–warm changes in KCE and gyre behavior, with variable expression among sites. M2 records the strongest cold-state response, with enhanced westerly/terrigeneous influence, gyre compression, and an equatorward KCE; warming into the mPWP reversed this pattern toward gyre expansion and a poleward KCE. Together, these results demonstrate spatial variability in bioevent timing and paleoceanographic responses across the Northwest Pacific.

Background and Objectives

Background

- Strong oceanographic gradients between the warm Kuroshio and cold Oyashio currents influence plankton ecology and distribution in the Northwest Pacific.
- ODP Sites 1207, 1208, and 1209 span subpolar/transitional to subtropical waters, providing a latitudinal transect across these contrasting water masses (Fig. 1).

- 1) **Biostratigraphic gap:** Planktic foraminiferal bioevents are diachronous within this region (Lam et al., 2022), but the magnitude and spatial patterns of calcareous nannofossil diachroneity, and how they compare with foraminifera, remain poorly constrained.

- 2) **Paleoceanographic gap:** Previous mPWP studies resolved broad KCE changes across the transect (Lam et al., 2021), but high-resolution (~3 kyr) reconstruction is limited to Site 1207 (Heo, 2023), leaving event-scale variability across the transect unresolved.

Objectives

- 1) **Bioevent Diachroneity:**

Quantify calcareous nannofossil bioevents' diachroneity across Shatsky Rise and compare its magnitude and spatial patterns with planktic foraminifera.

- 2) **KCE Paleoenvironmental Reconstruction:**

Reconstruct high-resolution KCE variability during the mPWP using foraminiferal stable isotopes and XRF geochemistry.

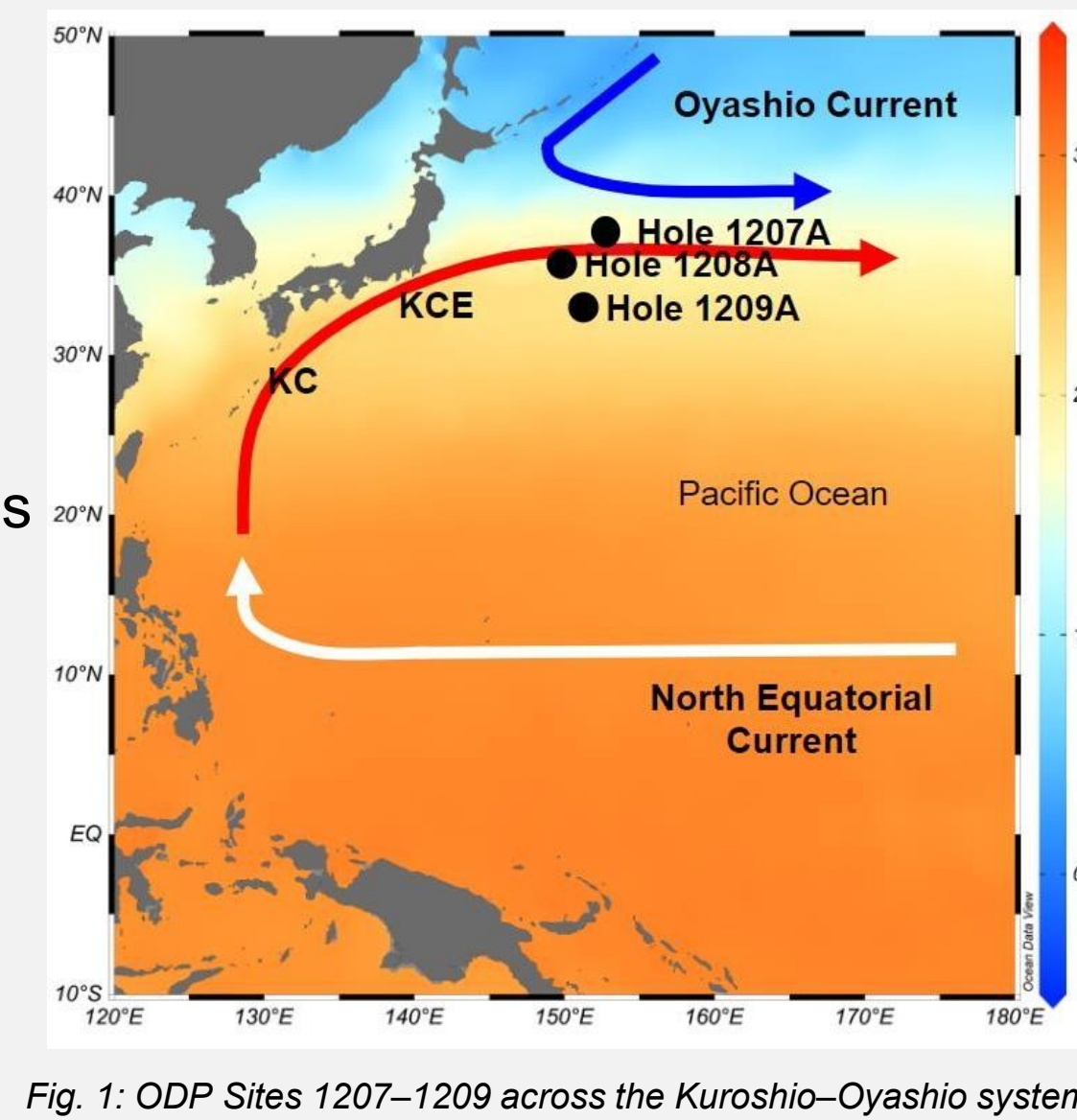


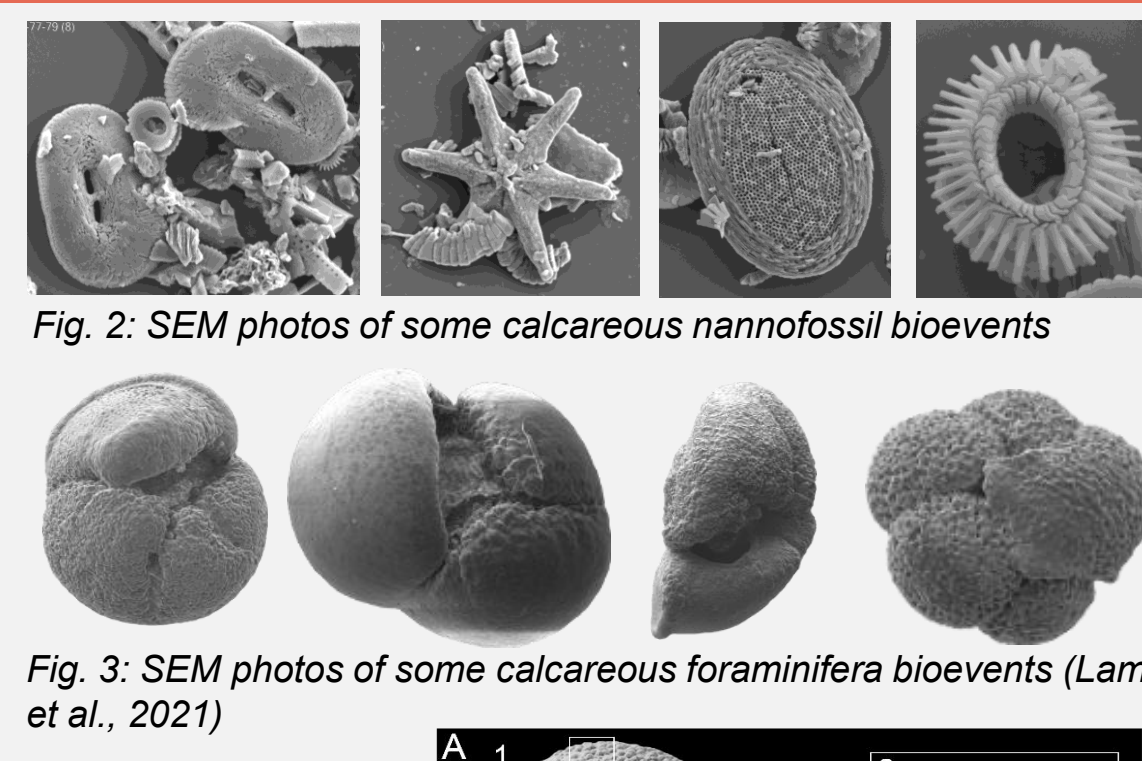
Fig. 1: ODP Sites 1207–1209 across the Kuroshio–Oyashio system

Methodology

1) Diachroneity

- Recalibrated nannofossil bioevents (Bralower et al., 2002; Bown, 2005) using independent age models tuned to GTS2020.

- Quantified diachroneity, compared magnitude and patterns with p. foraminifera.



2) Paleoenvironmental Reconstruction

- ~3-kyr foraminiferal (G. inflata; Figs. 4 & 5) $\delta^{18}\text{O}$ & $\delta^{13}\text{C}$; high-resolution XRF (Fig. 6) at 1207A–1209A.

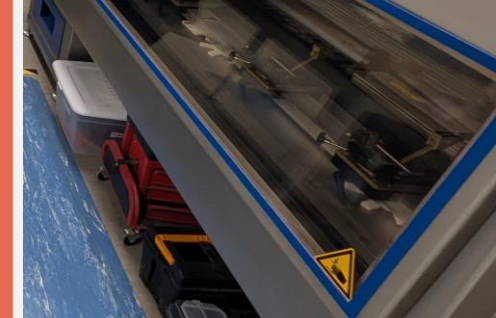


Fig. 6: XRF core scanning for high-resolution elemental analysis.

Fig. 5: G. inflata peaks in winter (Eguchi, 2003)

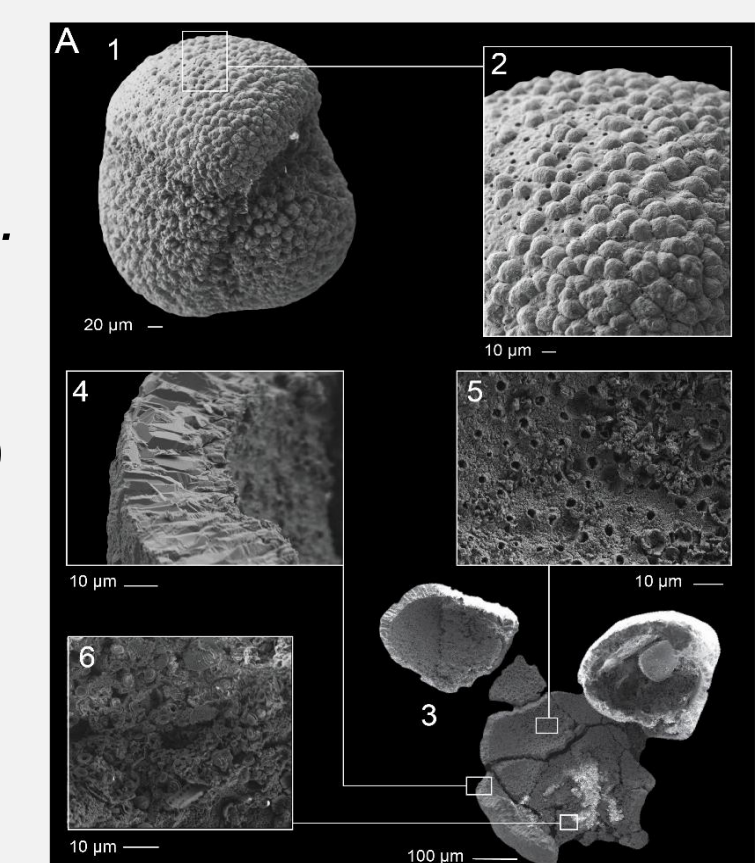


Fig. 4: G. inflata showing good preservation (Heo, 2023)

Results and Discussion

1) Magnitude of Bioevent Diachroneity in the Two Microfossil Groups

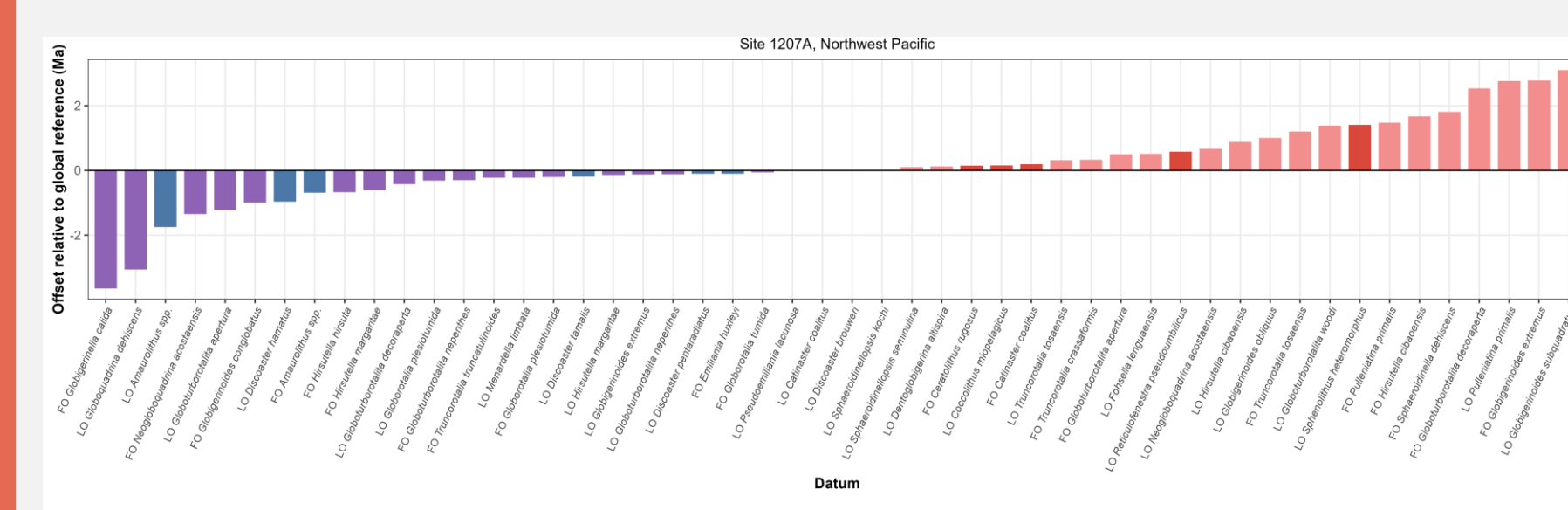


Fig. 7: Bioevent age offsets at Site 1207A.

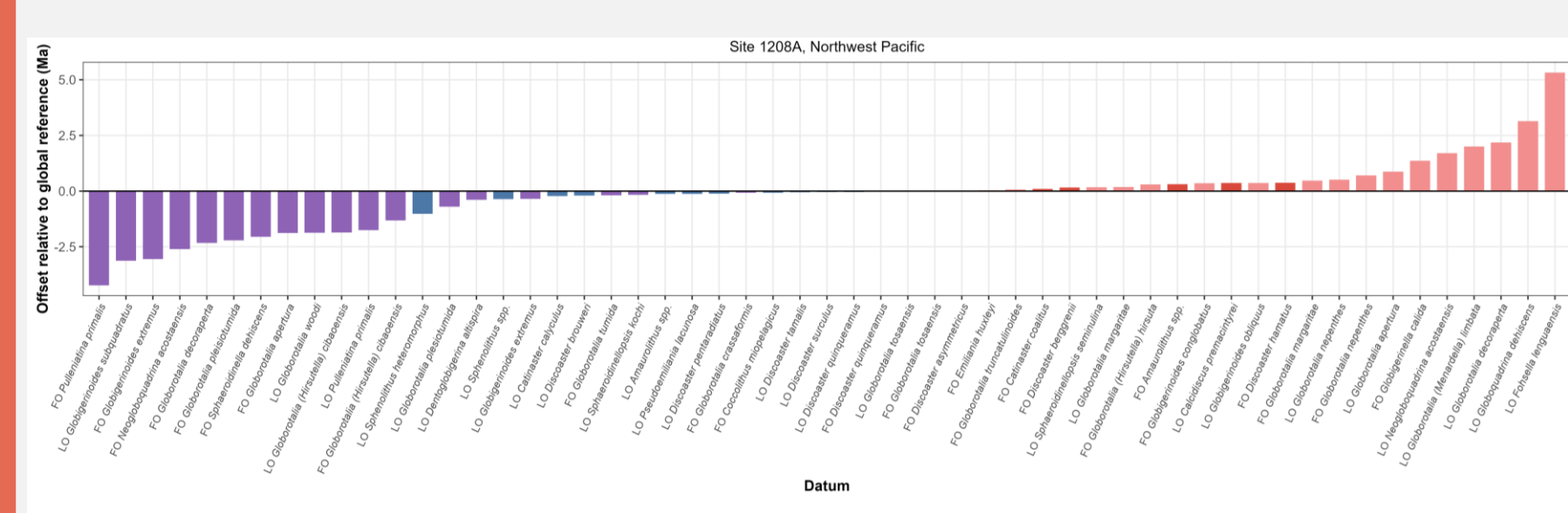


Fig. 8: Bioevent age offsets at Site 1208A.

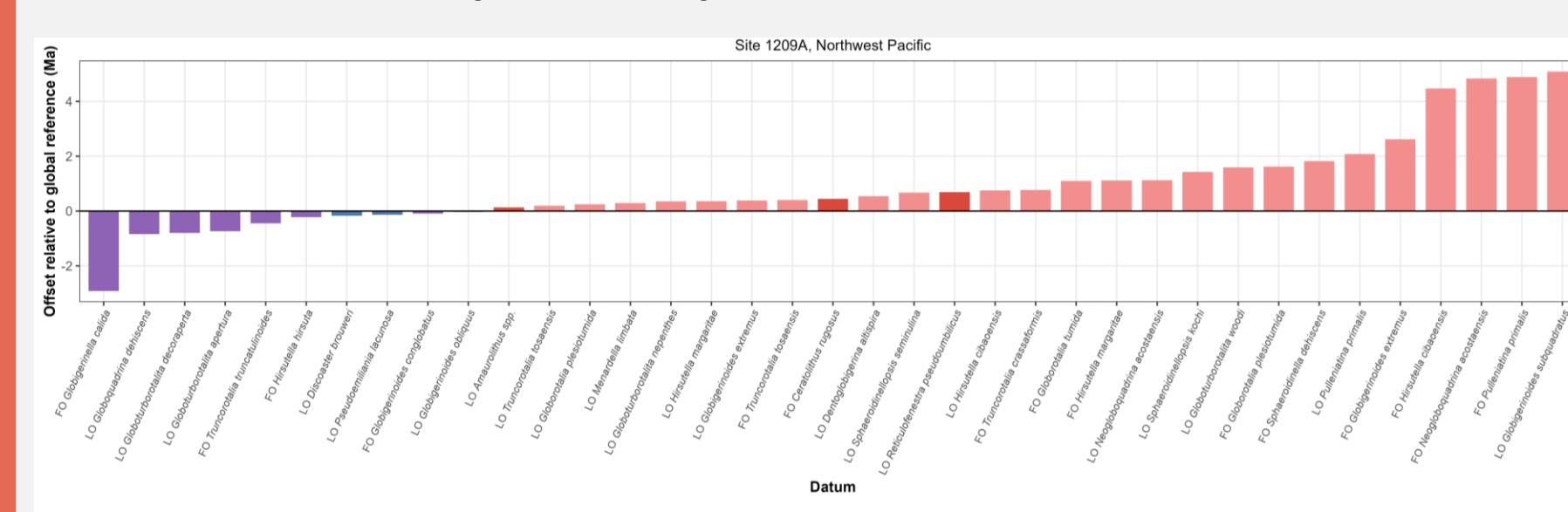


Fig. 9: Bioevent age offsets at Site 1209A.

- Both microfossil groups exhibit diachroneity, with nannofossil bioevents showing relatively smaller offsets (Figs. 7–10).

- Nannofossil LOs show a wider range of offsets than FOs, whereas foraminiferal FOs show a wider range than LOs (Fig. 11).

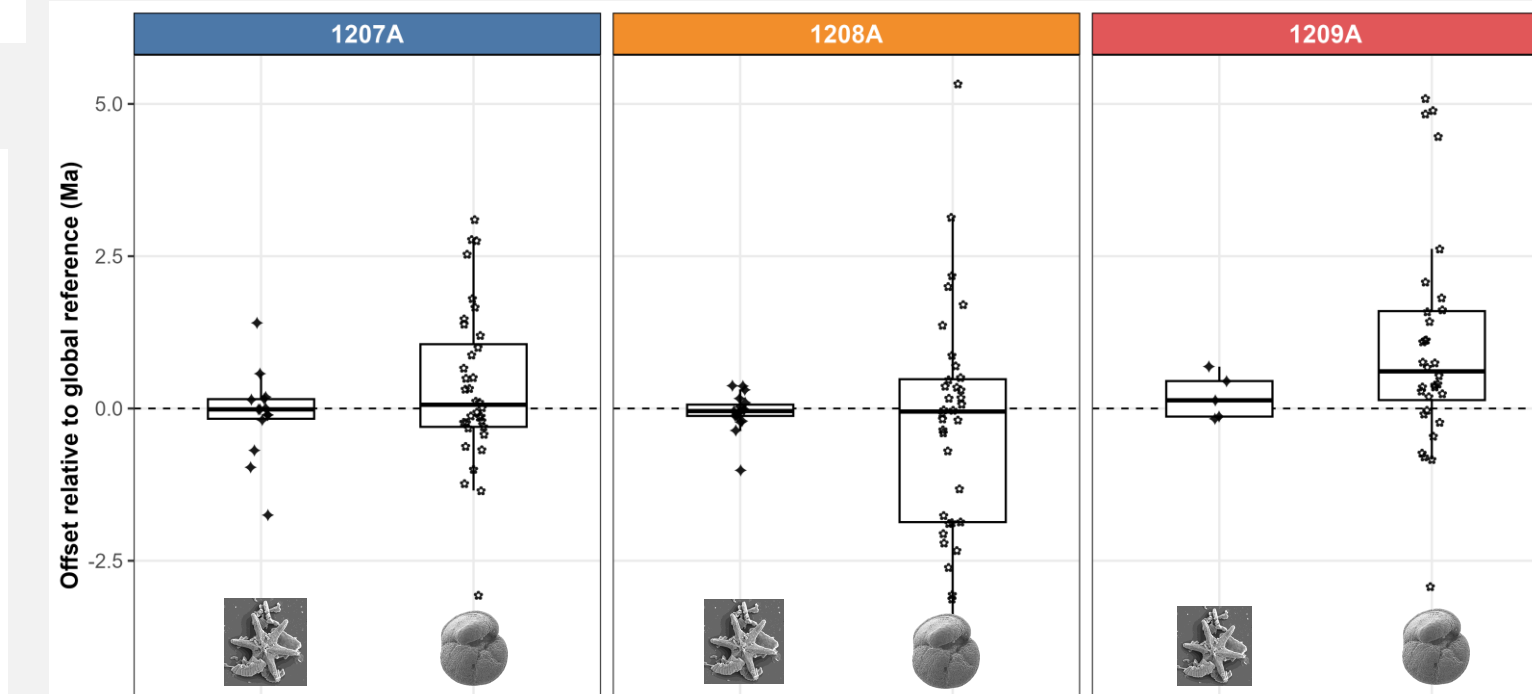


Fig. 11: Three-site summary of nannofossil vs. foraminiferal bioevents' offsets.

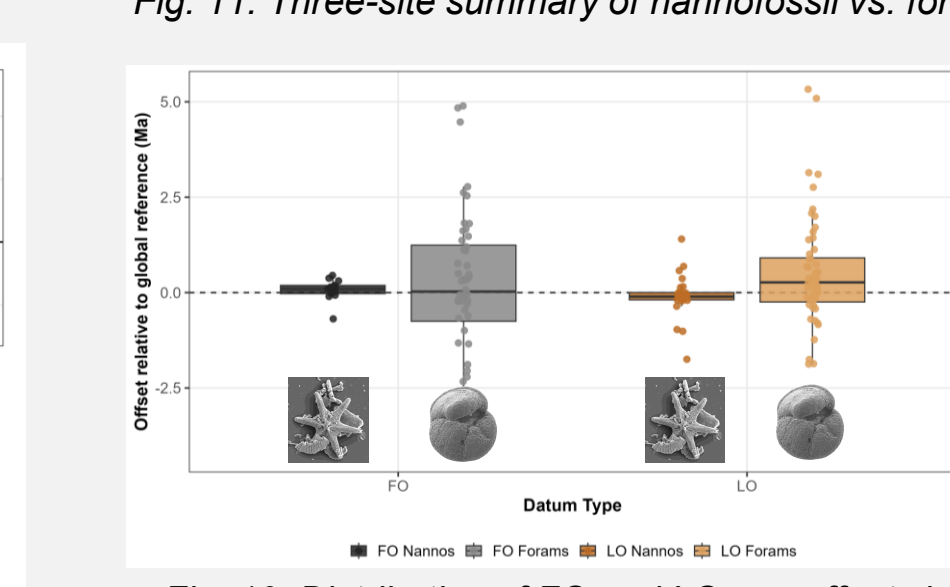


Fig. 10: Distribution of FO and LO age offsets in nannofossils and planktic foraminifera.

- Some diachronous events show systematic latitudinal patterns across the transect, others do not (Figs. 12–14).

Latitudinal Patterns of Bioevent Diachroneity



Fig. 12: Common nannofossil bioevents at all sites

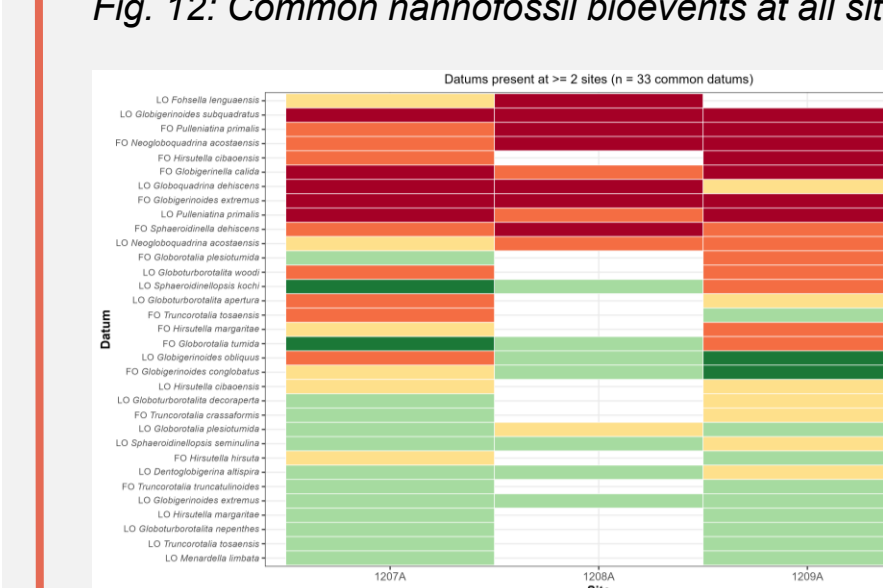


Fig. 13: Common p. foraminifera bioevents at all sites

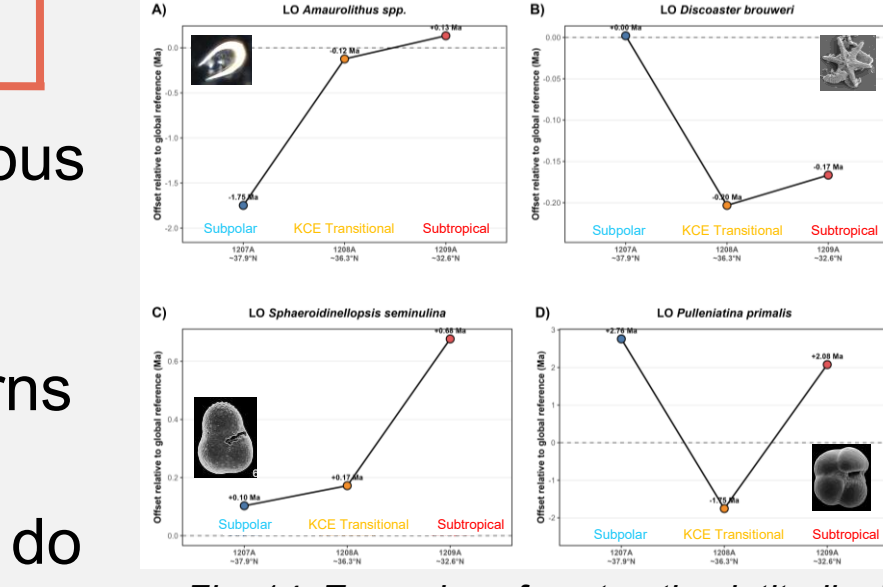


Fig. 14: Examples of contrasting latitudinal patterns in bioevent diachroneity.

2) Spatial–Temporal Thermocline Variability Across the KCE

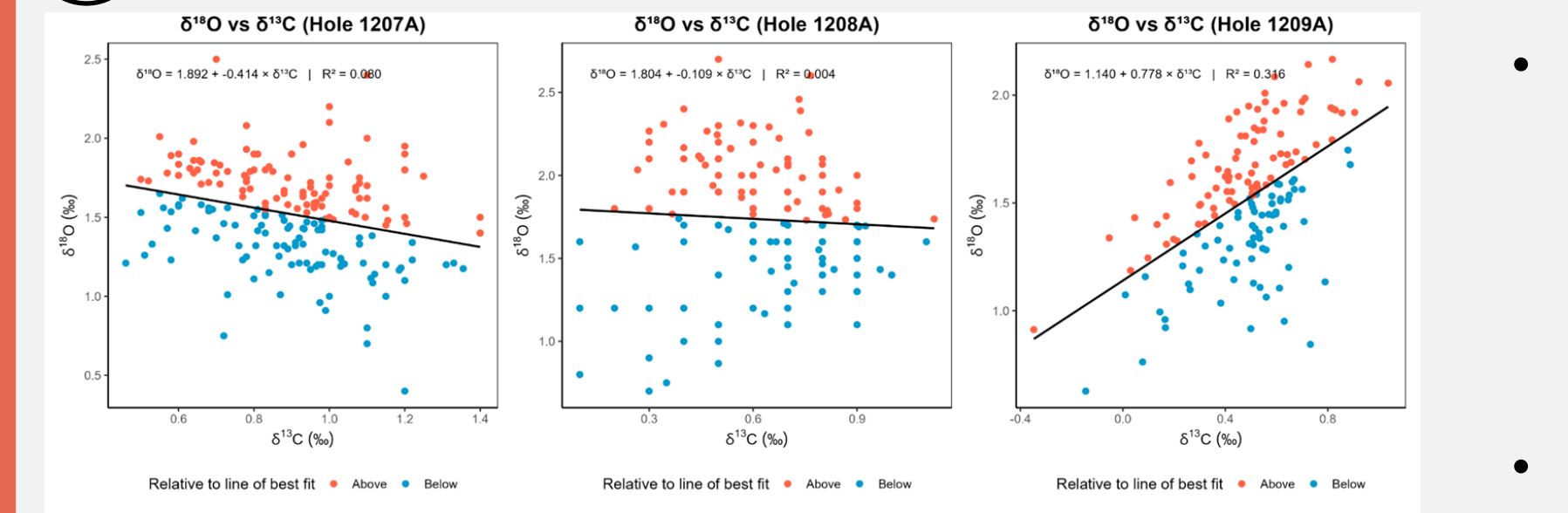


Fig. 15: 5°N–5°S Intra-Sample Variability across the sites

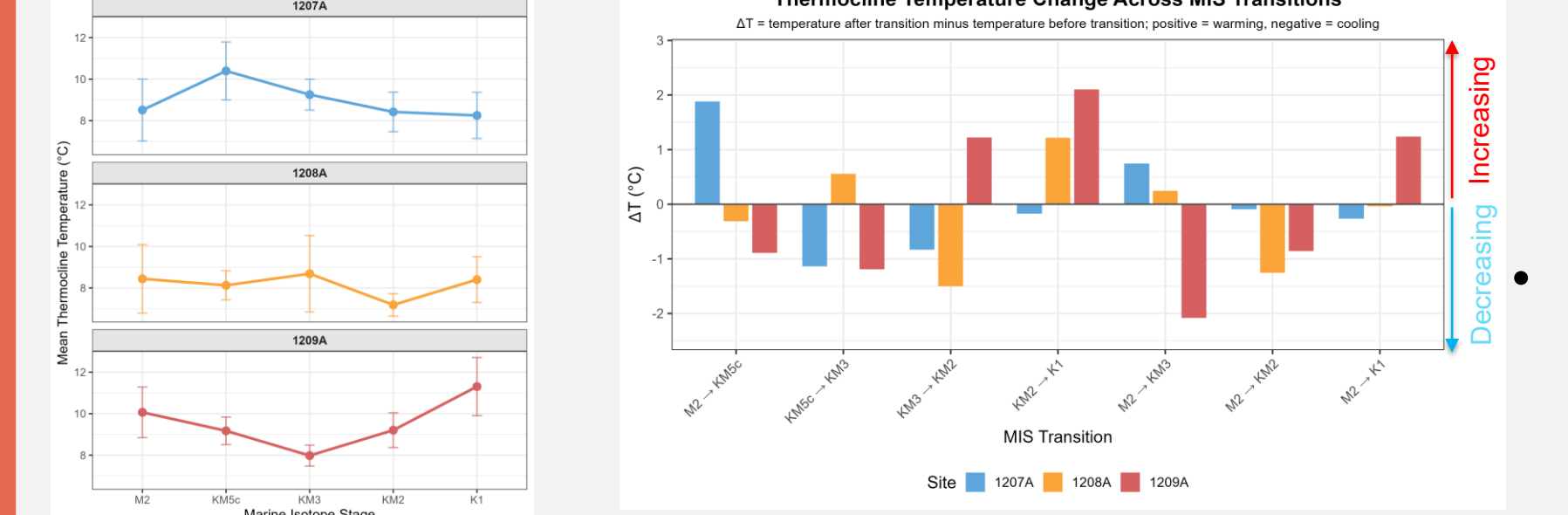


Fig. 16: Mean thermocline temperature change across MIS transitions

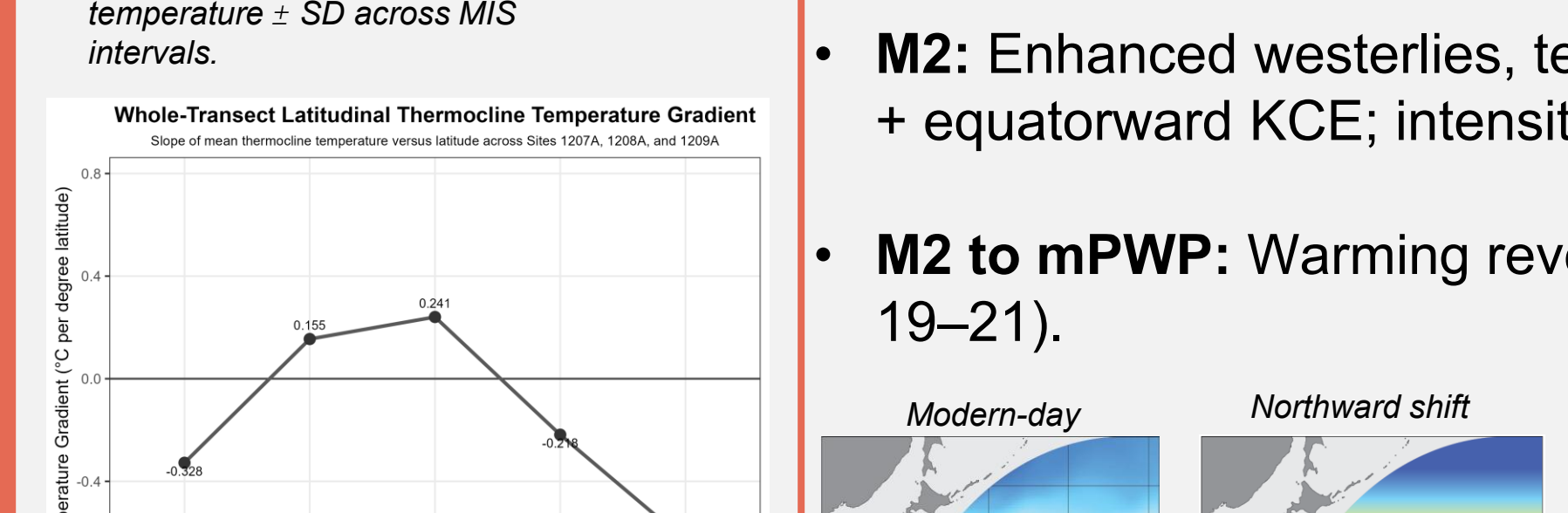


Fig. 17: Thermocline temperature change across MIS transitions



Fig. 18: Latitudinal thermocline temperature gradient across MIS intervals

Integrated Paleoenvironmental Reconstruction

- More variable water-mass conditions at 1207A and 1208A, near the KCE; more stable subtropical conditions at 1209A (Fig. 15).

- Thermocline temperatures varied among sites and MIS intervals, indicating spatially heterogeneous temperature conditions (Figs. 16 & 17).

- The north–south thermocline temperature gradient changed in both direction and magnitude among MIS intervals (Fig. 18).

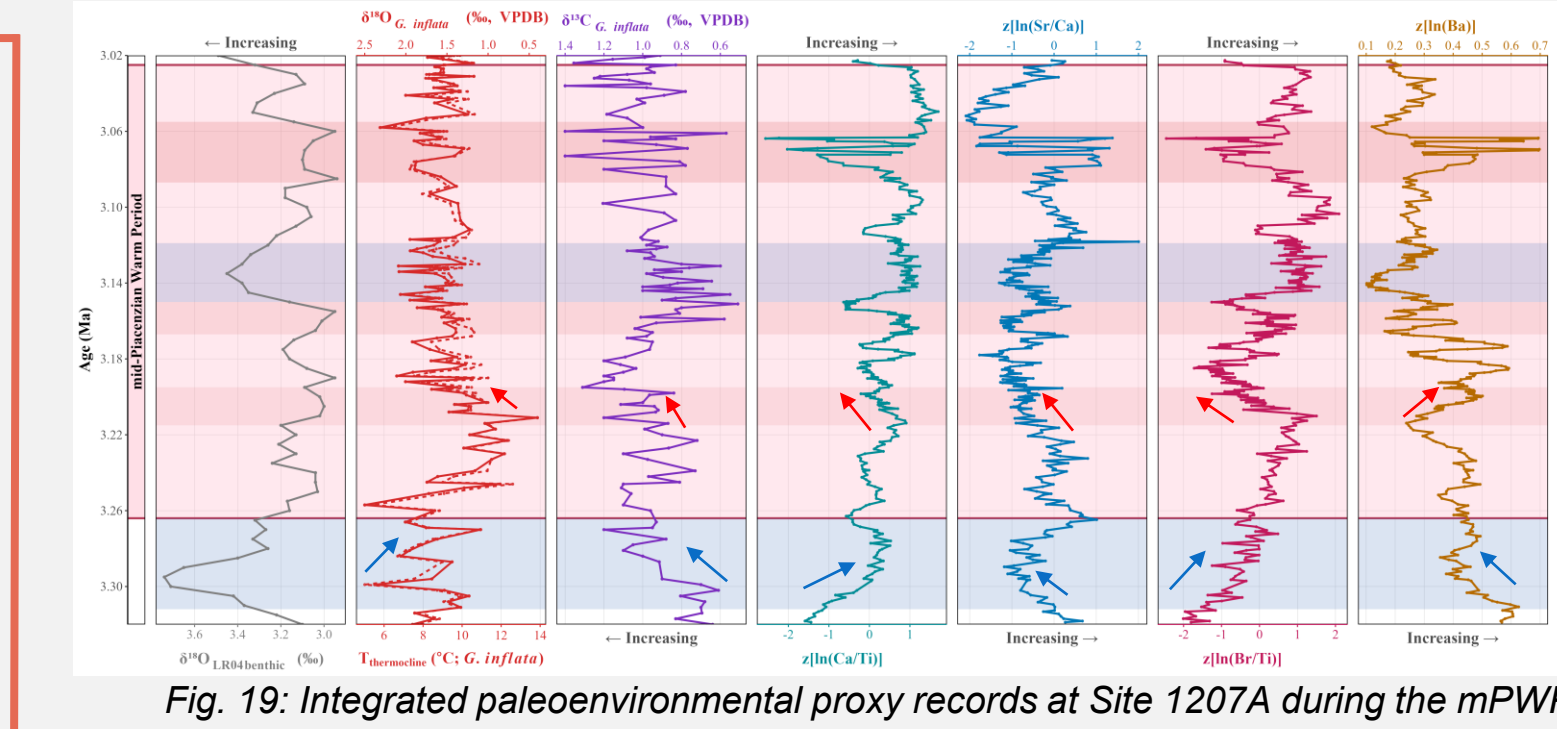


Fig. 19: Integrated paleoenvironmental proxy records at Site 1207A during the mPWP.

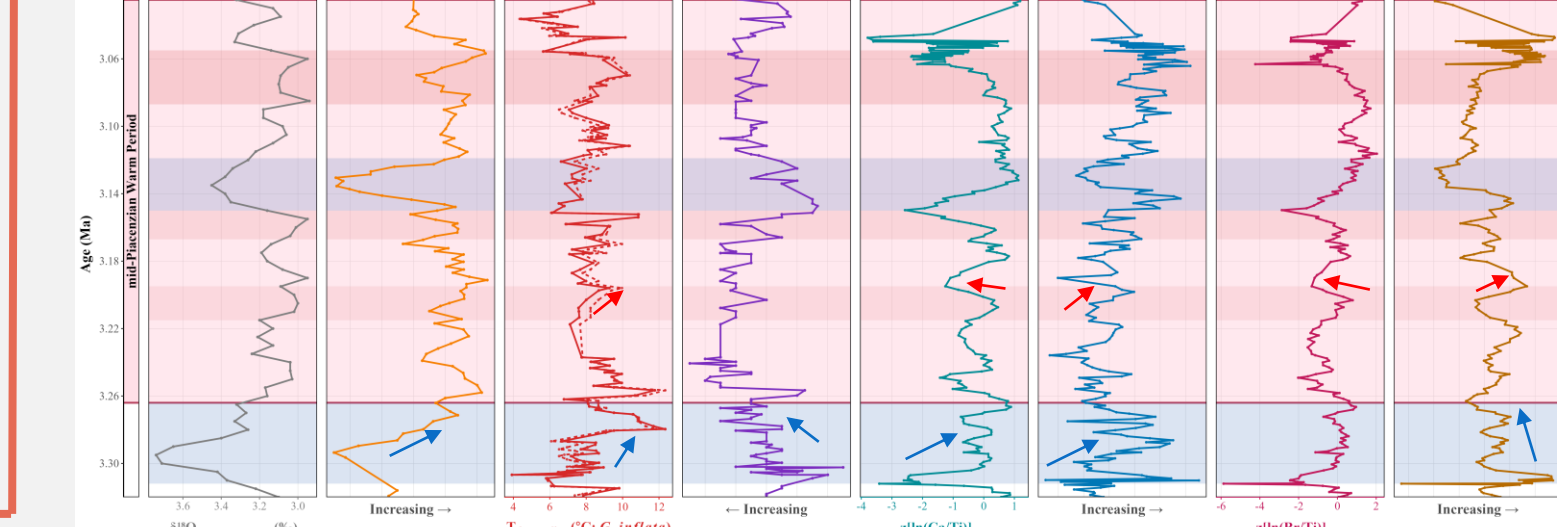


Fig. 20: Integrated paleoenvironmental proxy records at Site 1208A during the mPWP.

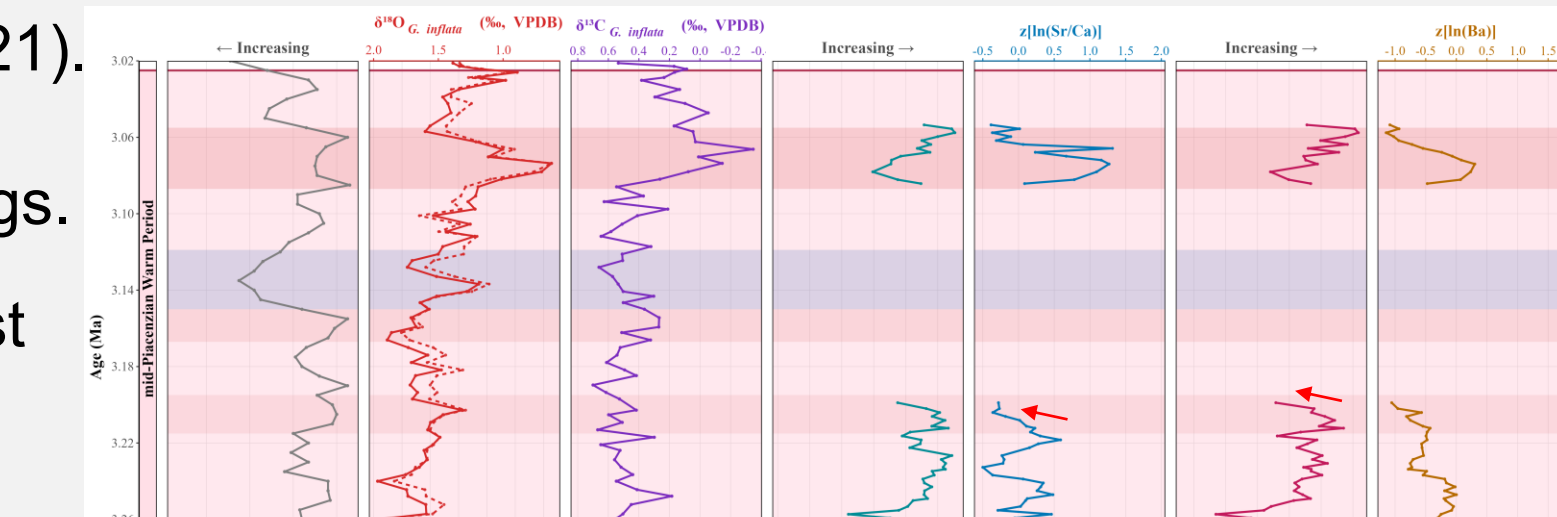
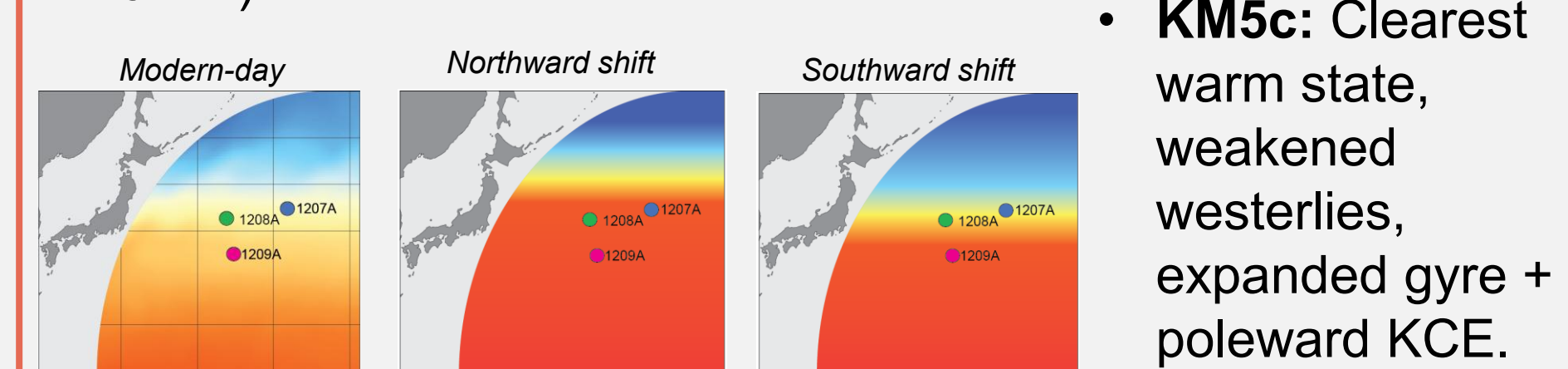


Fig. 21: Integrated paleoenvironmental proxy records at Site 1209A during the mPWP.

- M2:** Enhanced westerlies, terrigenous input, gyre compression + equatorward KCE; intensity varies among sites (Figs. 19–21).

- M2 to mPWP:** Warming reverses the pattern seen in M2 (Figs. 19–21).



- KM5c:** Clearest warm state, weakened westerlies, expanded gyre + poleward KCE.

Conclusions and Future Work

1) Diachroneity

- Bioevents in both microfossil groups exhibit varying magnitudes and spatial patterns of diachroneity, with nannofossils generally showing smaller offsets than planktic foraminifera.

- Our analysis has already been expanded across the entire western Pacific; next, we will evaluate basin-wide latitudinal patterns and possibly link them to associated drivers.

2) KCE Paleoenvironmental Reconstruction

- KCE conditions varied spatially and temporally during the mPWP, with climate transitions associated with shifts in KCE and gyre behavior and the influence of the westerlies across the transect.

- Assemblage work is ongoing to characterize water masses and will be integrated with existing proxies to strengthen the multi-proxy reconstruction (Figs. 22 & 23).

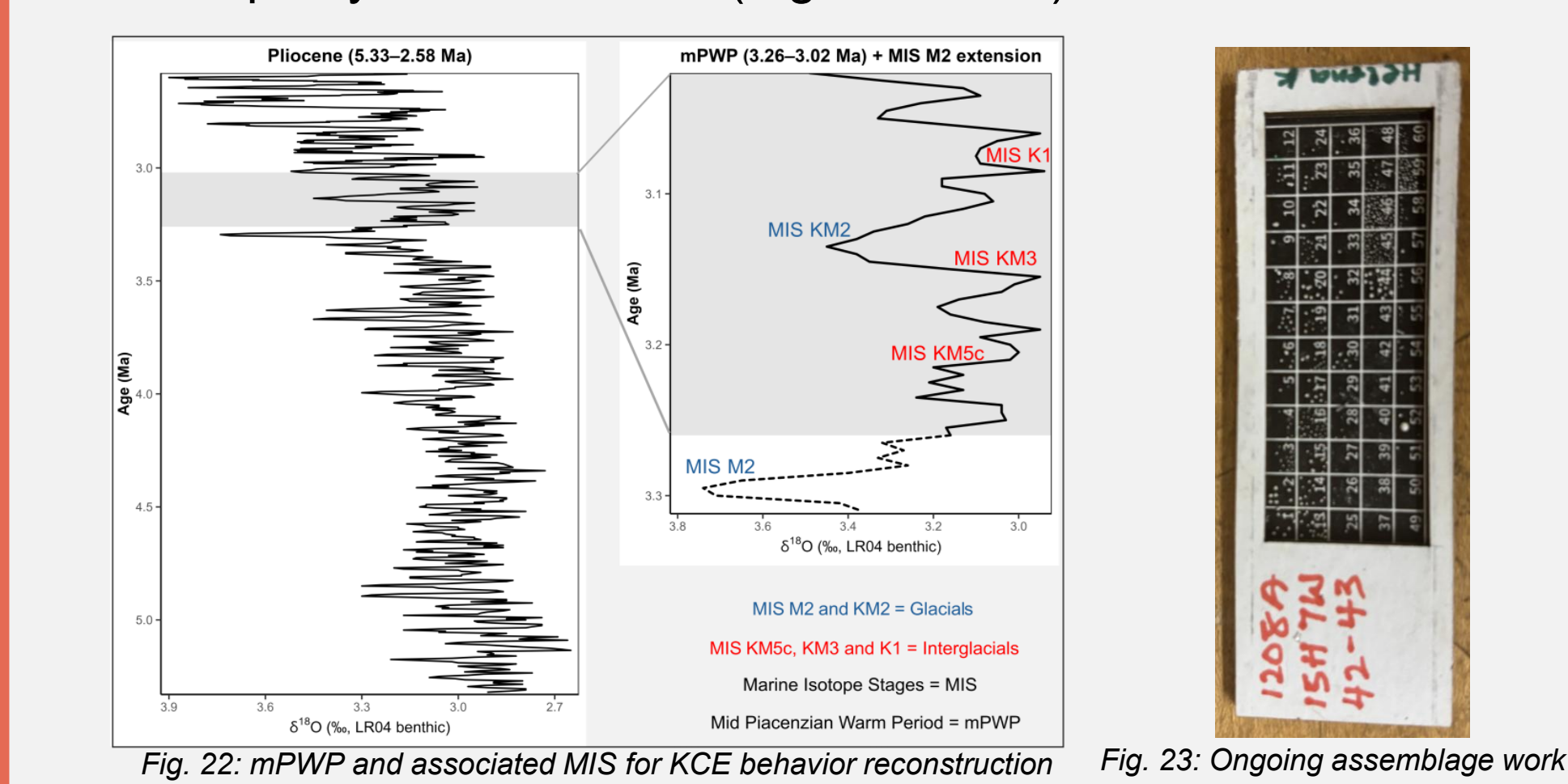
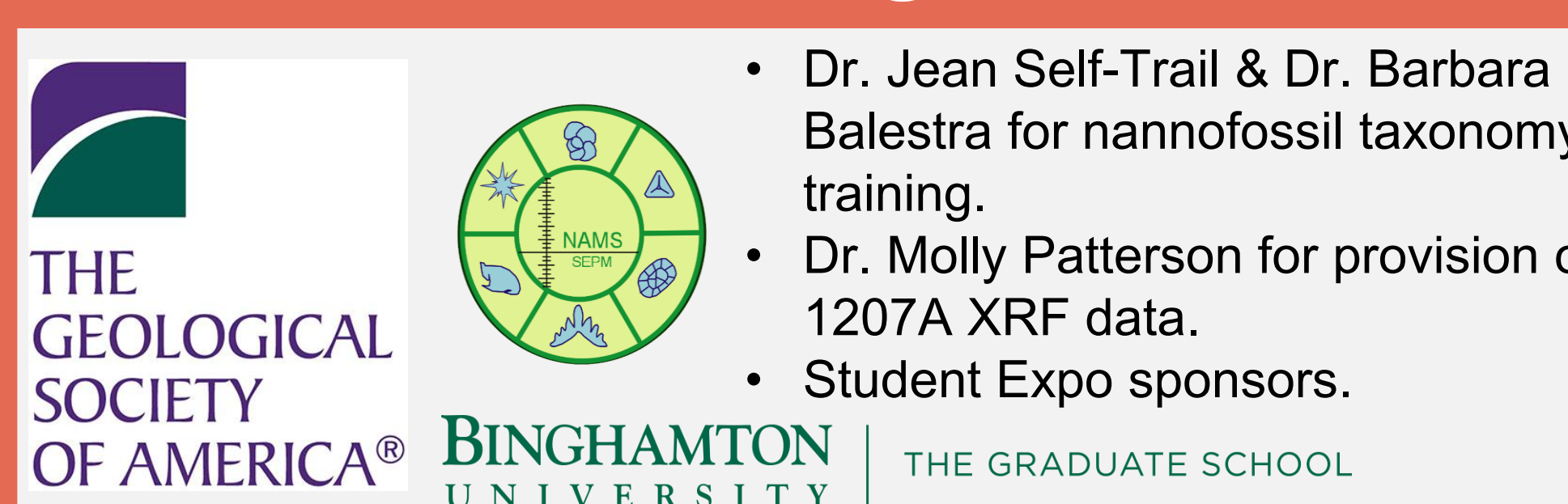


Fig. 22: mPWP and associated MIS for KCE behavior reconstruction



Fig. 23: Ongoing assemblage work

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